A sharp spectral-deficit stability inequality for triangle-free graphs

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Abstract

Let G be a finite simple triangle-free graph with m edges and adjacency spectral radius λ, and let $D(G)$ be the least number of edge insertions and deletions that turn G into a complete bipartite graph together with isolated vertices. Nosal's inequality states that $\lambda^2 \le m$. We prove that $D(G) \le 3(m - \lambda^2)$. The five-cycle attains equality, so the constant 3 is best possible. For bipartite graphs this inequality is a lemma of Li, Liu and Ning; the theorem removes the bipartiteness assumption. The proof averages the costs of the targets $K_{N(v),N(u)}$ over the edges uv of G with the weights $x_u^2/d(u) + x_v^2/d(v)$, where x is a unit eigenvector for λ, and bounds this average by two Cauchy–Schwarz inequalities and a counting lemma that uses triangle-freeness. The theorem and the sharpness example are formalized in Lean 4 with Mathlib; the proof is accepted by Lean's kernel and by five further proof checkers.

1 Introduction

For a finite simple graph G we write $e(G)$ for its number of edges and $\lambda(G)$ for the spectral radius of its adjacency matrix ($\lambda(G) = 0$ when G has no vertices). Nosal [7] proved that every triangle-free graph satisfies $\lambda(G)^2 \le e(G)$; see also Nikiforov [6]. Complete bipartite graphs attain equality, since $\lambda(K_{s,t})^2 = st = e(K_{s,t})$, and adding isolated vertices changes neither side. We measure the distance from this extremal situation by the *spectral deficit*

$$\xi(G) = e(G) - \lambda(G)^2 \;\ge\; 0,$$

and the distance from complete bipartite graphs by

$$D(G) = \min_{\substack{A,B \subseteq V(G) \\ A \cap B = \varnothing}} \bigl| E(G) \triangle E(K_{A,B}) \bigr|,$$

where $K_{A,B}$ is the graph on $V(G)$ whose edges are the pairs with one end in A and the other in B, and empty parts are allowed. Thus $D(G)$ is the least number of edge insertions and deletions that turn G into a complete bipartite graph together with isolated vertices (the graph without edges included).

Li, Liu and Zhang [4, Theorem 4.1] proved a qualitative edge-spectral stability theorem for triangle-free graphs. For bipartite graphs, Li, Liu and Ning [3, Lemma 4.3] proved the linear bound $D(H) \le 3\,\xi(H)$; their target is $K_{N(v),N(u)}$ for a suitable edge uv, found by averaging over all edges. The question whether the same coefficient holds for every triangle-free graph was posed in [2], together with the bound $D(G) \le 15\,\xi(G)$. We answer it.

Theorem 1. *For every finite simple triangle-free graph G,*

$$D(G) \;\le\; 3\big(e(G) - \lambda(G)^2\big).$$

The constant cannot be lowered. Label the five-cycle C_5 cyclically by $1, \dots, 5$. Then $e(C_5) = 5$ and $\lambda(C_5) = 2$, so $\xi(C_5) = 1$, and $D(C_5) = 3$: the target $K_{\{1\},\{2,5\}}$ costs the three edges 23, 34, 45, and no target costs less (Section 6). The balanced blow-ups $C_5[t]$ attain equality as well (Remark 7).

The proof gives slightly more: when G has an edge, there is an edge uv whose target $K_{N(v),N(u)}$ costs at most $3\,\xi(G)$, as in the bipartite lemma of [3]. Since $D(G) = 0$ only for complete bipartite graphs with isolated vertices, Theorem 1 also contains the equality case of Nosal's inequality: $\xi(G) = 0$ forces G to be such a graph.

Idea of the proof. For an edge uv of a triangle-free graph the neighbourhoods $N(u)$ and $N(v)$ are disjoint, so $K_{N(v),N(u)}$ is a target. Li, Liu and Ning average its cost uniformly over the edges, which works for bipartite graphs. For triangle-free graphs the uniform average can exceed 3ξ (Remark 5). We choose a vertex u with probability x_u^2, where x is a unit eigenvector for λ, and then a uniformly random neighbour v of u. Two Cauchy–Schwarz inequalities for $\|Ax\|^2 = \lambda^2$ and one counting lemma (Lemma 3), the only place where triangle-freeness is used beyond the disjointness of $N(u)$ and $N(v)$, show that the expected cost is at most 3ξ.

2 Notation and the edge targets

Let G be a triangle-free graph on the vertex set V with m edges and adjacency matrix A. We write $d(u)$ for the degree of u, $d = A\mathbf{1}$ for the degree vector, $N(u)$ for the neighbourhood of u and $N[u] = N(u) \cup \{u\}$. For $S \subseteq V$ and $z \in V$ let $d_S(z) = |N(z) \cap S|$. For a vertex u put

$$W_u = V \setminus N[u], \qquad Z_u = e\big(G[W_u]\big).$$

The codegree $c(u,z) = |N(u) \cap N(z)|$ equals $(A^2)_{uz}$, and $c(u,u) = d(u)$. Hence $(A^2 d)_u = \sum_z c(u,z)\,d(z)$ and $(A^4)_{uu} = \sum_z c(u,z)^2$. Triangle-freeness gives two facts that we use repeatedly: $N(u)$ is an independent set for every u, and $c(u,v) = 0$ for every edge uv.

Because $N(u)$ is independent, every edge that meets $N(u)$ meets it in exactly one end. An edge misses $N(u)$ exactly when both its ends lie in W_u, since u has no neighbour in W_u. Therefore

$$(Ad)_u = \sum_{v \in N(u)} d(v) = m - Z_u. \tag{1}$$

For an edge uv define the cost of the target $K_{N(v),N(u)}$,

$$D_{uv} = \big|E(G) \triangle E(K_{N(v),N(u)})\big|.$$

As $N(u) \cap N(v) = \varnothing$, this target is admissible, and $D(G) \le D_{uv}$.

Lemma 2. *For every edge uv, $D_{uv} = m + d(u)d(v) - 2(A^3)_{uv}$. For every vertex u with $d(u) \ge 1$,*

$$\overline{D}_u := \frac{1}{d(u)} \sum_{v \in N(u)} D_{uv} \;=\; 2m - Z_u - \frac{2(A^4)_{uu}}{d(u)}. \tag{2}$$

Proof. The target has $d(u)d(v)$ edges. An edge of G belongs to it exactly when it has one end in $N(u)$ and the other in $N(v)$. Since these sets are disjoint, each such edge is counted exactly once in $\sum_{a\in N(u)}\sum_{b\in N(v)} A_{ab} = (A^3)_{uv}$. Hence $D_{uv} = m + d(u)d(v) - 2(A^3)_{uv}$. Summing over $v \in N(u)$, and using $\sum_v A_{uv}(A^3)_{vu} = (A^4)_{uu}$ and (1),

$$\sum_{v\in N(u)} D_{uv} = d(u)\, m + d(u)\,(Ad)_u - 2(A^4)_{uu} = d(u)\,(2m - Z_u) - 2(A^4)_{uu}. \qquad \square$$

3 A counting lemma

Lemma 3. *For every vertex u with $d(u) \geq 1$,*

$$(A^2 d)_u - (A^4)_{uu} = \sum_{zz'\in E(G[W_u])} \big(d_{N(u)}(z) + d_{N(u)}(z')\big) \;\leq\; d(u)\, Z_u.$$

Proof. We have $(A^2 d)_u - (A^4)_{uu} = \sum_z c(u,z)\big(d(z) - c(u,z)\big)$. The term $z = u$ vanishes because $c(u,u) = d(u)$, and the terms $z \in N(u)$ vanish because $c(u,z) = 0$ for an edge uz. For $z \in W_u$ the vertex z is not adjacent to u, so $N(z) \subseteq N(u) \cup W_u$; hence $c(u,z) = d_{N(u)}(z)$ and $d(z) - c(u,z) = d_{W_u}(z)$. Therefore

$$(A^2 d)_u - (A^4)_{uu} = \sum_{z\in W_u} d_{N(u)}(z)\, d_{W_u}(z) = \sum_{zz'\in E(G[W_u])} \big(d_{N(u)}(z) + d_{N(u)}(z')\big),$$

where the second sum counts each edge zz' of $G[W_u]$ once from each end. For such an edge, triangle-freeness gives $N(z) \cap N(z') = \varnothing$, so $d_{N(u)}(z) + d_{N(u)}(z') = |(N(z) \cup N(z')) \cap N(u)| \leq d(u)$. Summing over the Z_u edges of $G[W_u]$ gives the bound. $\square$

The slack of the lemma is $\sum_{zz'\in E(G[W_u])} |N(u) \setminus (N(z) \cup N(z'))|$.

4 Two Cauchy–Schwarz bounds

Lemma 4. *Let G be a graph without isolated vertices and let $y \in \mathbb{R}^V$. Then*

$$y^{\mathsf{T}} A^2 y \leq \sum_u y_u^2\,(Ad)_u, \tag{3}$$

$$y^{\mathsf{T}} A^2 y \leq \sum_u y_u^2\,\frac{(A^2 d)_u}{d(u)}. \tag{4}$$

Proof. We have

$$y^{\mathsf{T}} A^2 y = \|Ay\|^2 = \sum_v \Big(\sum_{u\in N(v)} y_u\Big)^2.$$

For (3), bound $\big(\sum_{u\in N(v)} y_u\big)^2 \leq d(v)\sum_{u\in N(v)} y_u^2$ and exchange the order of summation:

$$\sum_v d(v) \sum_{u\in N(v)} y_u^2 = \sum_u y_u^2 \sum_{v\in N(u)} d(v) = \sum_u y_u^2\,(Ad)_u.$$

For (4), use the weights $d(u)$:

$$\Big(\sum_{u\in N(v)} y_u\Big)^2 \leq \Big(\sum_{u\in N(v)} d(u)\Big)\Big(\sum_{u\in N(v)} \frac{y_u^2}{d(u)}\Big) = (Ad)_v \sum_{u\in N(v)} \frac{y_u^2}{d(u)},$$

and exchange the order of summation: $\sum_v (Ad)_v \sum_{u\in N(v)} y_u^2/d(u) = \sum_u \big(y_u^2/d(u)\big) \sum_{v\in N(u)} (Ad)_v$, and $\sum_{v\in N(u)} (Ad)_v = (A^2 d)_u$. □

5 Proof of Theorem 1

If $m = 0$, then $D(G) = 0$ (take $A = B = \varnothing$) and $\lambda = 0$, so the statement holds. Let $m \geq 1$. Deleting the isolated vertices changes neither m nor λ. It does not change $D(G)$ either: an isolated vertex can be removed from either part of a target without increasing its cost, and every target of the smaller graph is a target of G. So we may assume that every vertex has degree at least one.

The adjacency matrix is symmetric and nonnegative, so its spectral radius λ is its largest eigenvalue. Let x be a real unit eigenvector for λ; any such vector works. Then $x^{\mathsf{T}} A^2 x = \|Ax\|^2 = \lambda^2$ and $\sum_u x_u^2 = 1$. Adding (3) and twice (4) at $y = x$, and bounding $(A^2 d)_u \leq (A^4)_{uu} + d(u) Z_u$ by Lemma 3,

$$3\lambda^2 \leq \sum_u x_u^2 \Big[(Ad)_u + \frac{2(A^4)_{uu}}{d(u)} + 2Z_u \Big] = \sum_u x_u^2 \Big[m + Z_u + \frac{2(A^4)_{uu}}{d(u)} \Big],$$

where the equality is (1). By (2),

$$\sum_u x_u^2 \, \overline{D}_u = \sum_u x_u^2 \Big[2m - Z_u - \frac{2(A^4)_{uu}}{d(u)} \Big] = 3m - \sum_u x_u^2 \Big[m + Z_u + \frac{2(A^4)_{uu}}{d(u)} \Big] \leq 3m - 3\lambda^2.$$

The left side is a convex combination of the numbers $\overline{D}_u$, and each $\overline{D}_u$ is the mean of D_{uv} over the nonempty set $N(u)$. Hence some vertex u has $\overline{D}_u \leq 3\xi$, and some neighbour v of u has $D_{uv} \leq \overline{D}_u \leq 3\xi$. Therefore $D(G) \leq D_{uv} \leq 3\xi(G)$. □

In terms of edges, the inequality just proved says that

$$\sum_{uv\in E(G)} \Big(\frac{x_u^2}{d(u)} + \frac{x_v^2}{d(v)} \Big) D_{uv} \;\leq\; 3\,\xi(G),$$

where the weights are nonnegative and sum to 1.

6 Sharpness

For the five-cycle, $m = 5$ and the all-ones vector is a positive eigenvector with eigenvalue 2, so $\lambda(C_5) = 2$ and $\xi(C_5) = 1$. For disjoint $A, B \subseteq V(C_5)$ with $a = |A|$ and $b = |B|$, the cost of $K_{A,B}$ is $5 + ab - 2e(A,B)$, where $e(A,B)$ is the number of edges between A and B. We show that $2e(A,B) - ab \leq 2$, which gives $D(C_5) \geq 3$. We may assume $a \leq b$, so $a \leq 2$ since $a + b \leq 5$. If $a = 0$ the claim is clear. If $a = 1$, then $e(A,B) \leq \min(2, b)$ and $2\min(2,b) - b \leq 2$. If $a = 2$, then $e(A,B) \leq 4$ because every vertex has degree two, and $e(A,B) \leq 3$ when $b = 2$ because C_5 has no four-cycle; so $2e(A,B) - ab \leq \max(6-4,\, 8-6) = 2$. The target $K_{\{1\},\{2,5\}}$ costs 3. Hence $D(C_5) = 3 = 3\,\xi(C_5)$, and no coefficient below 3 bounds D by a multiple of ξ on all triangle-free graphs.

7 Remarks

Remark 5 (Other weightings). The uniform average of D_{uv} over the edges, which proves the bipartite lemma of [3], can exceed 3ξ for triangle-free graphs. A search over all labelled triangle-free graphs

on at most seven vertices, with eigenvalues computed to 50 digits, finds no example on five or fewer vertices. On six vertices the largest excess occurs for the blow-up $C_5[1,1,2,1,1]$ of the five-cycle in which one vertex is replaced by two nonadjacent copies. There $m = 7$ and the seven edge costs are $5,5,5,3,3,3,3$, so the uniform average is $27/7$. The characteristic polynomial is $z(z^2+z-1)(z^3-z^2-5z+4)$, and λ is the largest root r of the cubic factor. The excess is $27/7 - 3(7-r^2) = 3r^2 - 120/7 \approx 0.0133$. It is positive because the cubic is increasing on $[2,\infty)$ and takes the value $-12/7 + 10\sqrt{70}/49 < 0$ at $\sqrt{40/7}$, so $r^2 > 40/7$.

Weights proportional to $x_u x_v$ also fail. In the double star $S(8,7)$, two adjacent vertices a and b with 8 and 7 pendant neighbours, the edge costs are 7 on the pendant edges at a, 8 on the pendant edges at b, and 56 on the edge ab. Here $\lambda^2 = 8 + 2\sqrt{2}$ and $3\xi = 24 - 6\sqrt{2}$, while $\frac{2}{\lambda}\sum_{uv\in E} x_u x_v D_{uv} = 15/2 + 6\sqrt{2}$, larger by $12\sqrt{2} - 33/2 \approx 0.47$. The bound $D \le 3\xi$ itself holds there with room, since a pendant edge at a gives a target of cost 7.

Remark 6 (A matrix form). Lemmas 3 and 4 show that for every triangle-free graph without isolated vertices the matrix

$$Q = \operatorname{diag}\Big(m + Z_u + \frac{2(A^4)_{uu}}{d(u)}\Big)_{u\in V} - 3A^2$$

is positive semidefinite; the key inequality of Section 5 is $x^{\mathsf{T}} Q x \ge 0$.

Remark 7 (Equality). The three inequalities used in Section 5 are all equalities for complete bipartite graphs and for the balanced blow-ups $C_5[t]$, in which every vertex of C_5 is replaced by an independent set of size t and every edge by a complete bipartite graph $K_{t,t}$. On $K_{s,t}$ the eigenvector is constant on each part and $G[W_u]$ has no edges. On the regular graph $C_5[t]$ the eigenvector is constant, and for every u each edge zz' of $G[W_u]$ joins the two classes at distance two from the class of u, whose neighbourhoods together contain $N(u)$. Hence the weighted average equals 3ξ on these graphs.

Equality in Theorem 1 itself also holds on every $C_5[t]$: $D(C_5[t]) = 3t^2 = 3\,\xi(C_5[t])$ for every $t \ge 1$. This follows from $\xi(H[t]) = t^2\xi(H)$ and $D(H[t]) = t^2 D(H)$, which hold for every graph H and every $t \ge 1$, where $H[t]$ replaces each vertex of H by an independent set of size t and each edge by a complete bipartite graph $K_{t,t}$. The first identity holds because $e(H[t]) = t^2 e(H)$ and the adjacency matrix of $H[t]$ is the Kronecker product of that of H with the all-ones $t \times t$ matrix, so its nonzero eigenvalues are t times those of H. For the second, write the cost of a target as $e(H[t]) - g(A,B)$ with the gain $g(A,B) = 2e(A,B) - |A||B|$. Fix A and B outside one class X, and let $a = |A \cap X|$ and $b = |B \cap X|$, so $a + b \le t$. Since X is independent, the gain has the form $\gamma + \alpha a + \beta b - ab$, which is at most $\gamma + t\max(\alpha,\beta,0)$; this value is attained by putting all of X into A, all of X into B, or X into neither part. Rounding the classes one at a time gives an optimal target that is a union of whole classes, and the gain of such a target is t^2 times the gain of the corresponding target of H.

8 Formal verification

Theorem 1 and the sharpness example are formalized in Lean 4 [1] with Mathlib [5] (Lean v4.35.0, Mathlib v4.35.0). The formal statement uses the following definitions, reproduced in Appendix A.

- `editCost G S T` is $|E(G) \triangle E(K_{S,T})|$ on the vertex type, and `editDistance G` is its minimum over all pairs of disjoint finite sets (S,T), empty sets allowed; this is $D(G)$.
- `spectralRadius G` is the largest absolute value of an eigenvalue of the real adjacency matrix (0 when there are no vertices). For an adjacency matrix this is the largest eigenvalue $\lambda(G)$.

- `G.CliqueFree 3` says that G has no triangle. The vertex type is any finite type with decidable equality, so disconnected graphs and isolated vertices are included.

Two theorems state the results.

- `TriangleFreeEdit.editDistance_le_three_deficit`
  For every such G, $(\texttt{editDistance G} : \mathbb{R}) \le 3\,(|E(G)| - \texttt{spectralRadius G}^2)$.

- `TriangleFreeSharpness.triangleFree_constant_sharp`
  For every real $c < 3$ some finite triangle-free graph has $D(G) > c\,\xi(G)$.

For both, the command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

The formal statement of the problem is a separate file, `Challenge.lean`, which contains the definitions and the theorem with the proof replaced by `sorry`. The tool `lake comparator` of Lean v4.35.0 checks that the proved theorem has the same statement as the challenge, uses only the permitted axioms, and is accepted by the kernel. With the option `--paranoid` it also replays the proof in five further kernels (Lean's paranoid kernel, `lean4lean`, `nanoda`, `con-leche` and `con-ron`); all six kernels accepted it on 25 October 2026. That run used the option `--inadvisably-no-sandbox`, because the sandbox requires Linux; a checker who does not trust our files can repeat it with the sandbox. The four external checkers are those bundled with the Lean v4.35.0 toolchain.

To reproduce the checks, run in the folder `lean/` of the repository (see Data availability): `lake exe cache get`, `lake build`, `lake env lean Audit.lean` and `lake comparator --paranoid`. The output of our run is the file `lean/checks/comparator-paranoid.txt`.

How the result was obtained

The problem was created by the math-conjecturer harness of exactory and posted on 22 October 2026 [2]. The proof was found on 24 October 2026 by the math-solver harness of exactory, in which the Anthropic language model Claude Fable 5.1 wrote the proof, Claude Opus 5.5 reviewed it independently, and the OpenAI model gpt-6-astra reviewed it adversarially. Every identity and inequality of the proof was also checked numerically, with exact integer arithmetic for the combinatorial steps, on 2,918 triangle-free graphs (the script `record/scripts/verify_proof_steps.py`, with its output in `record/checks/`). The Lean proof was written by OpenAI Codex with gpt-6-astra and checked as described above. No human mathematician has yet reviewed the proof.

Data availability

The Lean sources, the formal statement, the configuration of `lake comparator`, the logs of the checks and the scripts of the numerical checks are in the public repository https://github.com/exactory/exactory-math, folder `attempts/triangle-free-spectral-stability`. The version checked as described in this paper is commit `7194c5784dba0cfe43dd5951f65e62f1b9c9456e`: the Lean files are in `lean/`, the log of `lake comparator` in `lean/checks/`, and the scripts of the numerical checks of Remark 5 and of the proof steps, with their outputs, in `record/scripts/` and `record/checks/`.

References

[1] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in: *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[2] exactory, Sharp spectral-deficit stability for triangle-free graphs, open problem posted 22 October 2026, https://www.exactory.ai/math/5aa390a0-bf63-4326-9b97-b50c46334fd8.

[3] Y. Li, L. Liu and B. Ning, The spectral Erdős book theorem: sharp bounds and stability, arXiv:2609.20225v2.

[4] Y. Li, H. Liu and S. Zhang, An edge-spectral Erdős–Stone–Simonovits theorem and its stability, arXiv:2508.15271v2.

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[6] V. Nikiforov, On a theorem of Nosal, arXiv:2104.12171.

[7] E. Nosal, Eigenvalues of graphs, Master's thesis, University of Calgary, 1970.

A The formal statement

The definitions and the theorem statement of `Challenge.lean`, without its header comments. In `TriangleFreeEdit.lean` the same lines are followed by the proof.

```
namespace TriangleFreeEdit

variable {V : Type*} [Fintype V] [DecidableEq V]

def bicliqueEdges (S T : Finset V) : Finset (Sym2 V) :=
  (S ×ˢ T).image (fun p => s(p.1, p.2))

def editCost (G : SimpleGraph V) [DecidableRel G.Adj] (S T : Finset V) : ℕ :=
  (symmDiff G.edgeFinset (bicliqueEdges S T)).card

def disjointPairs (V : Type*) [Fintype V] [DecidableEq V] : Finset (Finset V × Finset V) :=
  (Finset.univ : Finset (Finset V × Finset V)).filter (fun p => Disjoint p.1 p.2)

lemma disjointPairs_nonempty : (disjointPairs V).Nonempty :=
  ⟨(∅, ∅), by simp [disjointPairs]⟩

lemma editCosts_nonempty (G : SimpleGraph V) [DecidableRel G.Adj] :
    ((disjointPairs V).image (fun p => editCost G p.1 p.2)).Nonempty :=
  Finset.Nonempty.image disjointPairs_nonempty _

noncomputable def editDistance (G : SimpleGraph V) [DecidableRel G.Adj] : ℕ :=
  ((disjointPairs V).image (fun p => editCost G p.1 p.2)).min' (by exact editCosts_nonempty G)

lemma adjMatrix_isHermitian (G : SimpleGraph V) [DecidableRel G.Adj] :
    (G.adjMatrix ℝ).IsHermitian := by
  exact Matrix.isHermitian_iff_isSymm.mpr (G.isSymm_adjMatrix)

noncomputable def spectralRadius (G : SimpleGraph V) [DecidableRel G.Adj] : ℝ :=
  if h : Nonempty V then
    (Finset.univ : Finset V).sup' Finset.univ_nonempty
      (fun i => |(adjMatrix_isHermitian G).eigenvalues i|)
```

```
  else 0

theorem editDistance_le_three_deficit (G : SimpleGraph V) [DecidableRel G.Adj]
    (hG : G.CliqueFree 3) :
    (editDistance G : ℝ) ≤ 3 * ((G.edgeFinset.card : ℝ) - (spectralRadius G) ^ 2) := by
  sorry

end TriangleFreeEdit
```